

ARIMA & SARIMA

05

Outline

- (15) Examples & overview
- (50) Non-seasonal ARIMA
 - In-class exercise
- (25) Seasonal ARIMA (SARIMA)
 - In-class exercise
- (25) Applications
 - ETS vs ARIMA
 - Warehouse location model
- Appendix: ARIMA theory*

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm}) \epsilon_t$$

Example: Corticosteroid drug cost in Australia

Demo

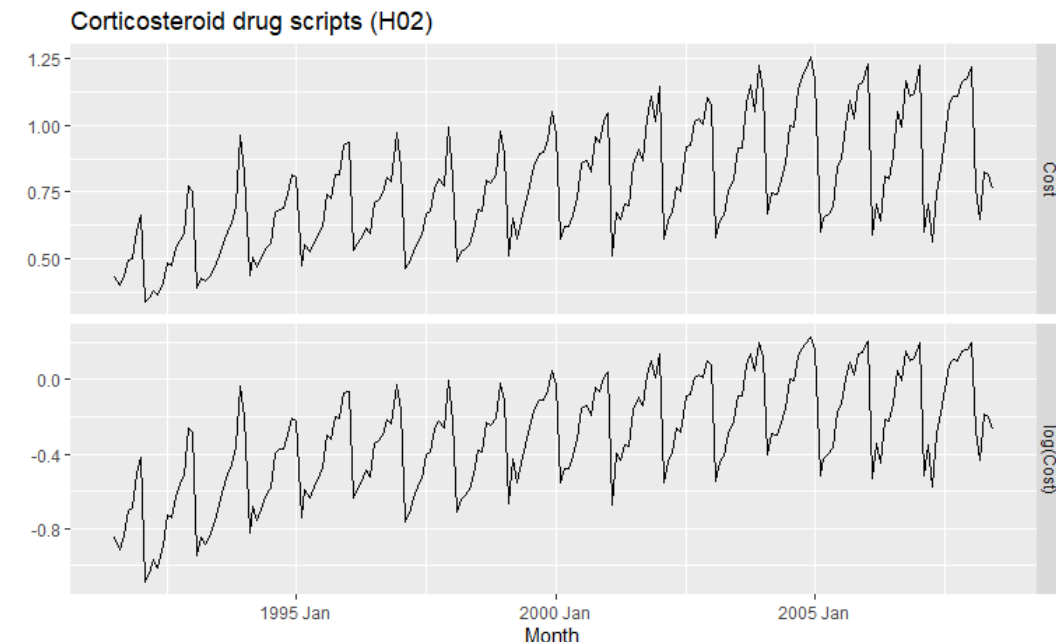
Monthly Medicare Australia prescription data (PBS)

2 values

- **Script** จำนวนใบสั่งยา (prescription)
- **Cost** ต้นทุนรวมการจ่ายยา (AUD) คิดรวมทั้ง co-payment ส่วนที่ผู้ป่วยจ่ายเอง & ส่วนที่รัฐบาลสนับสนุน

4 keys

- **ACT2** กลุ่มยา, e.g., ACT2 == "H02" กลุ่มยา Corticosteroids ใช้สำหรับภูมิแพ้ โรคหอบหืด. **Seasonal?**



```
h02 <- PBS %>%
  filter(ATC2 == "H02") %>%
  summarise(Cost = sum(Cost)/1e6)
fit <- h02 %>% model(auto = ARIMA(log(Cost)))
report(fit)
fit %>% qq_tsresiduals(lag_max=36)
```

```
> report(fit)
Series: Cost
Model: ARIMA(2,1,0)(0,1,1)[12]
Transformation: log(Cost)
```

```
Coefficients:
          ar1      ar2      sma1
        -0.8491  -0.4207  -0.6401
s.e.      0.0712   0.0714   0.0694
```

```
sigma^2 estimated as 0.004387: log likelihood=245.39
AIC=-482.78 AICc=-482.56 BIC=-469.77
```

We can write the model using the backshift notation as follows:

$$(1 - \phi_1 B - \phi_2 B^2)(1 - B)(1 - B^{12})y_t = c + (1 + \Theta_1 B^{12})\epsilon_t$$

$$(1 + 0.8491B + 0.4207B^2)(1 - B)(1 - B^{12})y_t = (1 - 0.6401B^{12})\epsilon_t$$

There is a small increase in the variance with the level, so we take logarithms to stabilize the variance.

ARIMA Overview

Autoregressive **i**ntegrated **m**oving **a**verage: **ARIMA**(**p**,**d**,**q**)

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d y_t = c + (1 + \theta_1 B^m + \dots + \theta_q B^q) \epsilon_t$$

ARIMA captures patterns (via **autocorrelations**) in TS by combining three ingredients:

AR (past values)	I (differences)	MA (past shocks)
AR order $p \in \mathbb{Z}_+$	d th differencing, usually 0,1,2	MA order $q \in \mathbb{Z}_+$

- In this class, we do not manually choose p , d , and q .
- Statistical software automatically selects an ARIMA model using criteria such as AIC, AICc, BIC, or likelihood.

Non-seasonal ARIMA(p,d,q)

Non-seasonal ARIMA(p,d,q) models

- Let ϵ_t be white noise, i.e., $\epsilon_t \sim \text{NID}(0, \sigma^2)$
- p = order of autoregressive part
- d = degrees of first differencing involved
- q = order of moving average part
- ARIMA model written using backshift notation

Note + or – sign in each book!

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{d^{\text{th}} \text{ order differences}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

ARIMA model written without using backshift notation

$$y_t^{(d)} = c + \phi_1 y_{t-1}^{(d)} + \phi_2 y_{t-2}^{(d)} + \dots + \phi_p y_{t-p}^{(d)} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

Non-seasonal ARIMA(p, d, q) models

$$\begin{array}{ccccc}
 (1 - \phi_1 B - \dots - \phi_p B^p) & (1 - B)^d y_t & = & c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t \\
 \uparrow & \uparrow & & \uparrow \\
 \text{AR}(p) & d \text{ differences} & & \text{MA}(q)
 \end{array}$$

- **Moving average (MA)** model of order q

$$\text{MA}(q): y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

- **Autoregressive (AR)** model of order p

$$\text{AR}(p): y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t$$

- **Autoregressive moving average (ARMA)** model

$$\text{ARMA}(p, q)$$

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- **Autoregressive integrated moving average ARIMA(p, d, q)**

$$y_t^{(d)} = c + \phi_1 y_{t-1}^{(d)} + \phi_2 y_{t-2}^{(d)} + \dots + \phi_p y_{t-p}^{(d)} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

where $y_t^{(d)}$ is the d th-order difference

For example $d = 1$, the differenced series is $y'_t = y_t - y_{t-1}$

$$y'_t = c + \phi_1 y'_{t-1} + \phi_2 y'_{t-2} + \dots + \phi_p y'_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

ARIMA(0,1,0): Random walk model

- Let ϵ_t be white noise, i.e., $\epsilon_t \sim \text{NID}(0, \sigma^2)$
- p = order of autoregressive part
- d = degrees of first differencing involved
- q = order of moving average part

- ARIMA model written using backshift notation

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t$$

AR(p)

dth order differences

MA(q)

Example: ARIMA(0,1,0) = Random walk model

Let $p = 0, d = 1, q = 0$. ARIMA(p,d,q) becomes

[p=0, d=1, q=0]

Expand differencing

Substitute backshift

Rearrange

- If the constant is zero, i.e., $c = 0$, this becomes a **random walk** $y_t = y_{t-1} + \epsilon_t$.
That is, the differenced series is white noise with zero mean: $y_t - y_{t-1} = \epsilon_t$.
 - **Naïve** method: today's freight rate = yesterday's rate + random fluctuation
- If $c \neq 0$, this becomes a **random walk with drift** $y_t = c + y_{t-1} + \epsilon_t$.
 - **Drift** method: today's cargo volume = steady growth + yesterday's volume + random fluctuation.

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{\text{d}^{\text{th}} \text{ order diff.}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

Non-seasonal ARIMA(p, d, q) models

Autoregressive (AR) models: ARIMA($p, 0, 0$)

$$AR(p): y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t$$

- AR means the current value depends linearly on *past values* of y 's, i.e., $y_{t-1}, y_{t-2}, \dots$
- Ex: Warehouse backlog today may depend on yesterday's backlog

AR(1): short memory	AR(2): longer memory
Ex: Today depends mainly on yesterday	Ex: Today depends on yester and two days ago.

Example

For an AR(1) model:

- when $\phi_1 = 0$ and $c = 0$, y_t is equivalent to white noise;
- when $\phi_1 = 1$ and $c = 0$, y_t is equivalent to a random walk;

AR(1):

$$(1 - \phi_1 B) y_t = c + \epsilon_t$$

$$y_t = c + \phi_1 y_{t-1} + \epsilon_t$$

AR(2):

$$(1 - \phi_1 B - \phi_2 B^2) y_t = c + \epsilon_t$$

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$$

AR(3):

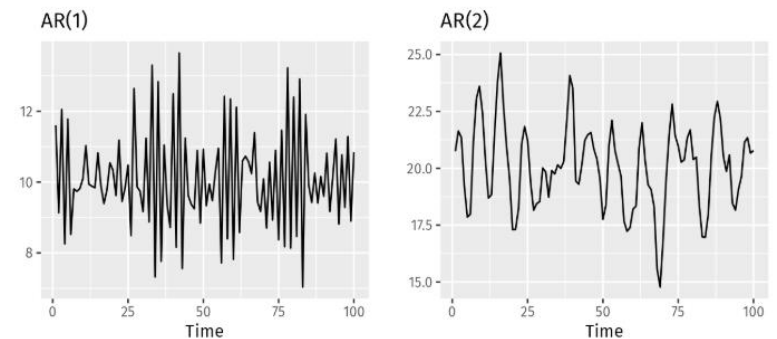


Figure 9.5: Two examples of data from autoregressive models with different parameters. Left: AR(1) with $y_t = 18 - 0.8y_{t-1} + \epsilon_t$. Right: AR(2) with $y_t = 8 + 1.3y_{t-1} - 0.7y_{t-2} + \epsilon_t$. In both cases, ϵ_t is normally distributed white noise with mean zero and variance one.

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{\text{d}^{\text{th}} \text{ order diff.}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

Non-seasonal ARIMA(p, d, q) models

Moving average (MA) models: [ARIMA\(0,0,q\)](#)

$$\text{MA}(q): y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

For an AR model, the past values of the forecast variable $y_{t-1}, y_{t-2}, \dots$ is used in a regression.

For an MA model, the past forecast errors (**shocks**) $\epsilon_{t-1}, \epsilon_{t-2}$ are used.

Ex: If a one-day disruption caused demand to be much higher than expected, an MA model allows that shock to affect the next forecast.

Example

$$\text{MA}(2): y_t = c + (1 + \theta_1 B + \theta_2 B^2) \epsilon_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2}$$

MA(4):

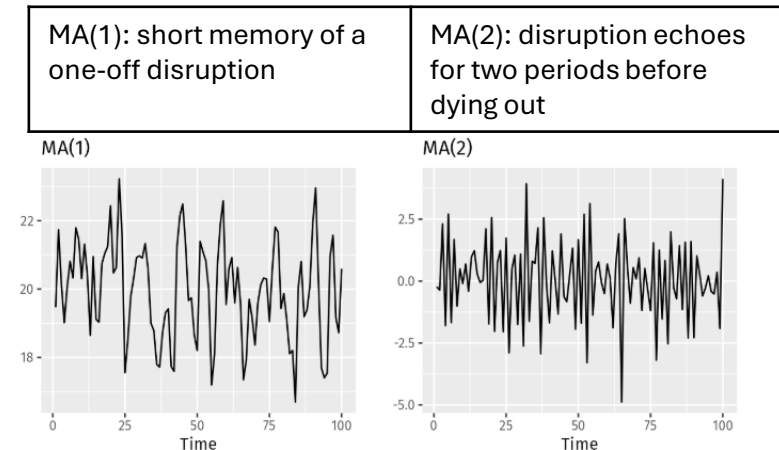


Figure 9.6: Two examples of data from moving average models with different parameters. Left: MA(1) with $y_t = 20 + \epsilon_t + 0.8\epsilon_{t-1}$. Right: MA(2) with $y_t = \epsilon_t - \epsilon_{t-1} + 0.8\epsilon_{t-2}$. In both cases, ϵ_t is normally distributed white noise with mean zero and variance one.

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{\text{d}^{\text{th}} \text{ order diff.}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

Non-seasonal ARIMA(p, d, q) models

Autoregressive moving average (**ARMA**) models

ARMA(p, q)

$$(1 - \phi_1 B - \dots - \phi_p B^p) y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t$$

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- ARMA combines two types of memory
- AR: memory of levels
- MA: memory of shocks

Example ARMA(2,1)

Warehouse backlog today may depend on:

- the backlog in the last two days; and
- yesterday's unexpected disruption.

Example

ARMA(1,1):

$$(1 - \phi_1 B) y_t = c + (1 + \theta_1 B) \epsilon_t$$

$$y_t = c + \phi_1 y_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

ARMA(2,1):

Non-seasonal ARIMA(p, d, q) models

$$\begin{array}{ccccc} (1 - \phi_1 B - \dots - \phi_p B^p) & (1 - B)^d y_t & = & c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t \\ \uparrow & \uparrow & & \uparrow \\ \text{AR}(p) & d \text{ differences} & & \text{MA}(q) \end{array}$$

- Moving average (MA) models

$$\text{MA}(q): \quad y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

- Autoregressive (AR) models

$$\text{AR}(p): \quad y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t$$

- Autoregressive moving average (ARMA) models

$$\text{ARMA}(p, q)$$

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- **ARIMA**(p, d, q)

$$y_t^{(d)} = c + \phi_1 y_{t-1}^{(d)} + \phi_2 y_{t-2}^{(d)} + \dots + \phi_p y_{t-p}^{(d)} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

where $y_t' = y_t - y_{t-1}$ is the differenced series (it may have been differenced more than once).

- p = order of autoregressive part
- d = degrees of first differencing involved
- q = order of moving average part

Example

$$\text{ARIMA}(1,1,1): \quad y_t' = c + \phi_1 y_{t-1}' + \theta_1 \epsilon_{t-1} + \epsilon_t$$

ARMA(1,1):

$$\begin{aligned} (1 - \phi_1 B) y_t &= c + (1 + \theta_1 B) \epsilon_t \\ y_t &= c + \phi_1 y_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t \end{aligned}$$

Example ARIMA(1,1,1)

$$\begin{array}{ccccc}
 (1 - \phi_1 B - \dots - \phi_p B^p) & (1 - B)^d y_t & = & c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t & (8.2) \\
 \uparrow & \uparrow & & \uparrow & \\
 \text{AR}(p) & d \text{ differences} & & \text{MA}(q) &
 \end{array}$$

ARIMA(1,1,1) in backshift notation

$$(1 - \phi_1 B)(1 - B)y_t = c + (1 + \theta_1 B)\epsilon_t$$

$$(1 - B - \phi_1 B + \phi_1 B^2)y_t = c + \epsilon_t + \theta_1 B \epsilon_t$$

$$y_t - y_{t-1} - \phi_1 y_{t-1} + \phi_1 y_{t-2} = c + \epsilon_t + \theta_1 \epsilon_{t-1} \quad \text{ARIMA(1,1,1) without using backshift notation}$$

$$(y_t - y_{t-1}) - \phi_1 (y_{t-1} - y_{t-2}) = c + \epsilon_t + \theta_1 \epsilon_{t-1}$$

$$y'_t - \phi_1 y'_{t-1} = c + \epsilon_t + \theta_1 \epsilon_{t-1}$$

$$y'_t = c + \phi_1 y'_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{\text{d}^{\text{th}} \text{ order diff.}} y_t = \underbrace{c + (1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

Non-seasonal ARIMA(p, d, q) models

- Moving average (MA) models

$$\text{MA}(q): y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}$$

- Autoregressive (AR) models

$$\text{AR}(p): y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \epsilon_t$$

- Autoregressive moving average (ARMA) models

$$\text{ARMA}(p, q)$$

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

- **ARIMA**(p, d, q)

$$y_t^{(d)} = c + \phi_1 y_{t-1}^{(d)} + \phi_2 y_{t-2}^{(d)} + \dots + \phi_p y_{t-p}^{(d)} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

where $y_t' = y_t - y_{t-1}$ is the differenced series (it may have been differenced more than once).

- p = order of autoregressive part
- d = degrees of first differencing involved
- q = order of moving average part

Example

$$\text{ARIMA}(1,1,1): y_t' = c + \phi_1 y_{t-1}' + \theta_1 \epsilon_{t-1} + \epsilon_t$$

$$\text{ARIMA}(1,2,1): y_t'' = c + \phi_1 y_{t-1}'' + \theta_1 \epsilon_{t-1} + \epsilon_t$$

ARIMA(1,1,1) warehouse throughput

- Trend $d = 1$: Throughput is growing over time (e.g., more e-commerce orders each year).
- AR $p = 1$: throughput changes have short persistence (momentum); e.g., big inbound shipment yesterday keeps warehouse busy into today.
- MA $q = 1$: one-off disruptions (e.g., conveyor belt broke down) spill into the next day.

Non-seasonal ARIMA

The following R code selects a non-seasonal ARIMA model automatically.

```
fit <- global_economy |>
  filter(Code == "EGY") |>
  model(ARIMA(Exports))
report(fit)
#> Series: Exports
#> Model: ARIMA(2,0,1) w/ mean
#>
#> Coefficients:
#>          ar1          ar2          ma1  constant
#>          1.6764    -0.8034    -0.6896      2.5623
#> s.e.    0.1111    0.0928    0.1492    0.1161
#>
#>           $\epsilon_t \sim NID(0, \sigma^2)$  with  $\hat{\sigma} = \sqrt{8.046} = 2.837$ 
#> sigma^2 estimated as 8.046: log likelihood=-141.6
#> AIC=293.1  AICc=294.3  BIC=303.4
```

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{\text{d}^{\text{th}} \text{ order diff.}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

We can write the model using the backshift notation as follows:

$$(1 - \phi_1 B - \phi_2 B^2) y_t = c + (1 + \theta_1 B) \epsilon_t$$

After substituting the estimated parameters, we get

$$(1 - 1.6764B + 0.8034B^2) y_t = 2.5623 + (1 - 0.6896B) \epsilon_t$$

Without the backshift notation, the model is

$$y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} = c + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Note the + and - signs

After substituting the estimated parameters, we get

$$y_t - 1.6764 y_{t-1} + 0.8034 y_{t-2} = 2.5623 + \epsilon_t - 0.6896 \epsilon_{t-1}$$

That is,

$$y_t = 2.5623 + 1.6764 y_{t-1} - 0.8034 y_{t-2} - 0.6896 \epsilon_{t-1} + \epsilon_t$$

- AR(2): reflect a 2-year boom–bust cycle (production, investment, or policy adjustments).
- MA(1): last year's disruptions (e.g., weather, Suez blockage, policy shocks) still echo into this year but fade quickly.
- ARMA(2,1): exports show cyclical ups and downs, with occasional one-year disruptions.

$$\underbrace{(1 - \phi_1 B - \dots - \phi_p B^p)}_{\text{AR}(p)} \underbrace{(1 - B)^d}_{\text{d}^{\text{th}} \text{ order diff.}} y_t = c + \underbrace{(1 + \theta_1 B + \dots + \theta_q B^q)}_{\text{MA}(q)} \epsilon_t$$

In-class exercise

Write the ARIMA model

- Using backshift notation
- Without backshift notation*

```
fit2 <- global_economy |>
  filter(Code == "EGY") |>
  model(ARIMA(Exports ~ pdq(4,1,2)))
report(fit2)
```

```
Series: Exports
Model: ARIMA(4,1,2)

Coefficients:
          ar1          ar2          ar3          ar4          ma1          ma2
          1.3557    -0.5902    0.2561    -0.2987    -1.3721    0.4356
s.e.    0.2766    0.3433    0.2287    0.1530    0.2743    0.2735

sigma^2 estimated as 8.183:  log likelihood=-138.66
AIC=291.32  AICc=293.61  BIC=305.63
```

ARIMA Notation: Two forms*

- Intercept form $(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q) \varepsilon_t$
- Mean form $(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d (y_t - \mu t^d / d!) = (1 + \theta_1 B + \dots + \theta_q B^q) \varepsilon_t$
- These two models are equivalent given that $c = \mu(1 - \phi_1 - \dots - \phi_p)$ and μ is the mean of $(1 - B)^d y_t$
- R uses mean form
- `fable` (also `fpp3`) uses intercept form

Forecasting with ARIMA models

```
fit <- global_economy |>
  filter(Code == "EGY") |>
  model(ARIMA(Exports))
report(fit)
#> Series: Exports
#> Model: ARIMA(2,0,1) w/ mean
#>
#> Coefficients:
#>      ar1      ar2      ma1  constant
#>      1.6764 -0.8034 -0.6896    2.5623
#> s.e.  0.1111  0.0928  0.1492    0.1161
#>
#> sigma^2 estimated as 8.046: log likelihood=-141.6
#> AIC=293.1  AICc=294.3  BIC=303.4
```

This is an ARIMA(2,0,1) model:

$$y_t = 2.56 + 1.68y_{t-1} - 0.80y_{t-2} - 0.69\varepsilon_{t-1} + \varepsilon_t,$$

```
fit |> forecast(h=10) |>
  autoplot(global_economy) +
  labs(y = "% of GDP", title = "Egyptian exports")
```

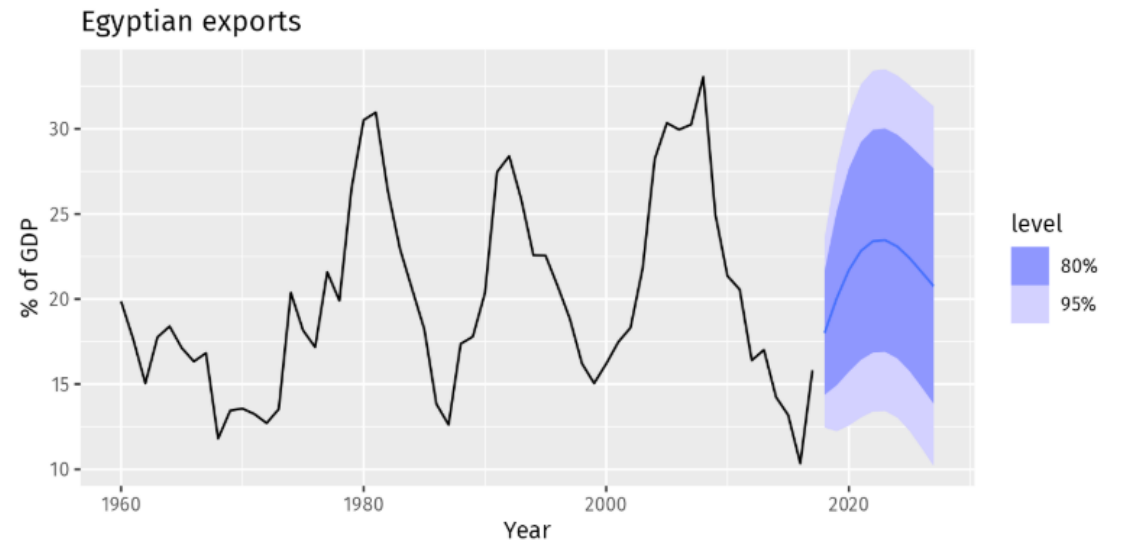


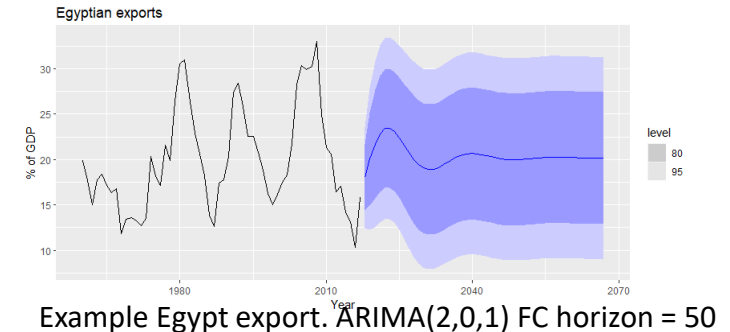
Figure 9.8: Forecasts of Egyptian exports.

Understanding ARIMA models

The constant c has an important effect on the long-term forecasts obtained from these models.

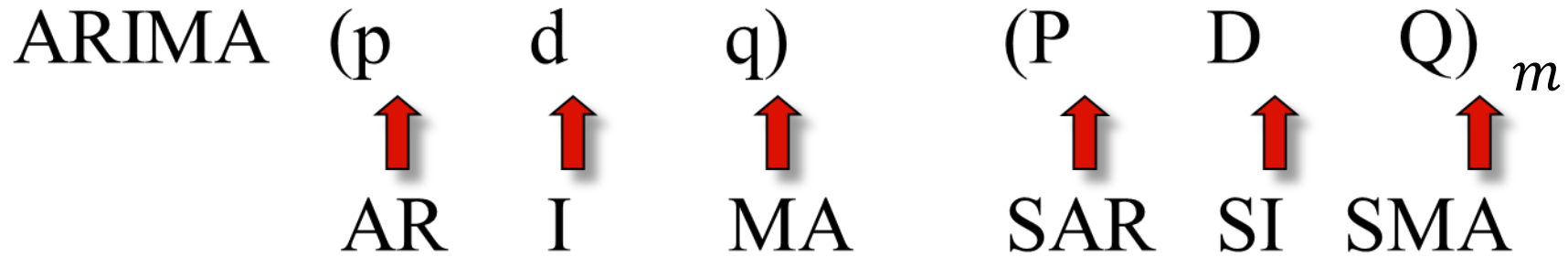
- If $c = 0$ and $d = 0$, the long-term forecasts will go to zero.
- If $c = 0$ and $d = 1$, the long-term forecasts will go to a non-zero constant.
- If $c = 0$ and $d = 2$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 0$, the long-term forecasts will go to the mean of the data.
- If $c \neq 0$ and $d = 1$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 2$, the long-term forecasts will follow a quadratic trend. (This is not recommended, and `fable` will not permit it.)

The value of d also has an effect on the prediction intervals — the higher the value of d , the more rapidly the prediction intervals increase in size. For $d = 0$, the long-term forecast standard deviation will go to the standard deviation of the historical data, so the prediction intervals will all be essentially the same.



Seasonal ARIMA (SARIMA)

Seasonal ARIMA Model



p is the number of AR terms,

d is the number of differences,

q is the number of MA terms,

P is the number of seasonal AR terms,

D is the number of seasonal differences,

Q is the number of seasonal MA terms.

m is the number of observations per year (periods per season).

Seasonal ARIMA(p, d, q)(P, D, Q) $_m$

p = non-seasonal AR order
 d = non-seasonal differencing
 q = non-seasonal MA order
 P = seasonal AR order
 D = seasonal differencing
 Q = seasonal MA order
 m = time span of repeating seasonal pattern (seasonal period)

$$\phi_p(B)\Phi_P(B^m)(1-B)^d(1-B^m)^D y_t = c + \theta_q(B)\Theta_Q(B^m)\epsilon_t$$

The non-seasonal components are

$$\text{AR: } \phi_p(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

$$\text{MA: } \theta_q(B) = 1 + \theta_1 B + \dots + \theta_q B^q$$

The seasonal components are

$$\text{Seasonal AR: } \Phi_P(B^m) = 1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm}$$

$$\text{Seasonal MA: } \Theta_Q(B^m) = 1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm}$$

That is,

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm})\epsilon_t$$

where ϵ_t is a white noise process

Seasonal ARIMA(p, d, q)(P, D, Q) $_m$

p = non-seasonal AR order
 d = non-seasonal differencing
 q = non-seasonal MA order
 P = seasonal AR order
 D = seasonal differencing
 Q = seasonal MA order
 m = time span of repeating seasonal pattern (seasonal period)

$$\phi_p(B)\Phi_P(B^m)(1-B)^d(1-B^m)^D y_t = c + \theta_q(B)\Theta_Q(B^m)\epsilon_t$$

The non-seasonal components are

$$\text{AR: } \phi_p(B) = 1 - \phi_1 B - \dots - \phi_p B^p$$

$$\text{MA: } \theta_q(B) = 1 + \theta_1 B + \dots + \theta_q B^q$$

The seasonal components are

$$\text{Seasonal AR: } \Phi_P(B^m) = 1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm}$$

$$\text{Seasonal MA: } \Theta_Q(B^m) = 1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm}$$

That is,

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm})\epsilon_t$$

where ϵ_t is a white noise process

Quarterly data: Seasonal period $m = 4$

$$P = 5, \Phi_P(B^m) = \Phi_5(B^4) = 1 - \Phi_1 B^4 - \Phi_2 B^8 - \Phi_3 B^{12} - \Phi_4 B^{16} - \Phi_5 B^{20}$$

$$Q = 3, \Theta_Q(B^m) =$$

Monthly data: Seasonal period $m = 12$

$$P = 3, \Phi_P(B^m) =$$

$$Q = 6, \Theta_Q(B^m) =$$

Seasonal AR vs Seasonal MA

Seasonal AR (Φ_p terms)

- *Air cargo peaks every year before Lunar New Year.*
- Structural, repeating pattern: this year's peak is correlated with last year's peak (not because of a shock, but because the seasonal cycle is built-in).
- Formula piece: $y_t = \dots + \Phi_1 y_{t-m} + \epsilon_t$.

Seasonal MA (Θ_Q terms)

- *Flood in last September damaged durian trees and still influences this year's harvest.*
- A one-off shock that repeats its effect at the seasonal lag. It's not the season itself, but the residual effect of last year's disruption that shapes this year's outcome.
- Formula piece: $y_t = \dots + \Theta_1 \epsilon_{t-m} + \epsilon_t$.

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm})\epsilon_t$$

Seasonal ARIMA

Example: ARIMA(1,1,1)(1,1,1)₄

- Backshift notation

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = c + (1 + \theta_1 B)(1 + \Theta_1 B^4)\epsilon_t$$

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm})\epsilon_t$$

Seasonal ARIMA

Example: ARIMA(**1**,1,1)(**1**,1,1)₄

- Backshift notation

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = c + (1 + \theta_1 B)(1 + \Theta_1 B^4)\epsilon_t$$

Non-seasonal AR(1)

Seasonal AR(1)

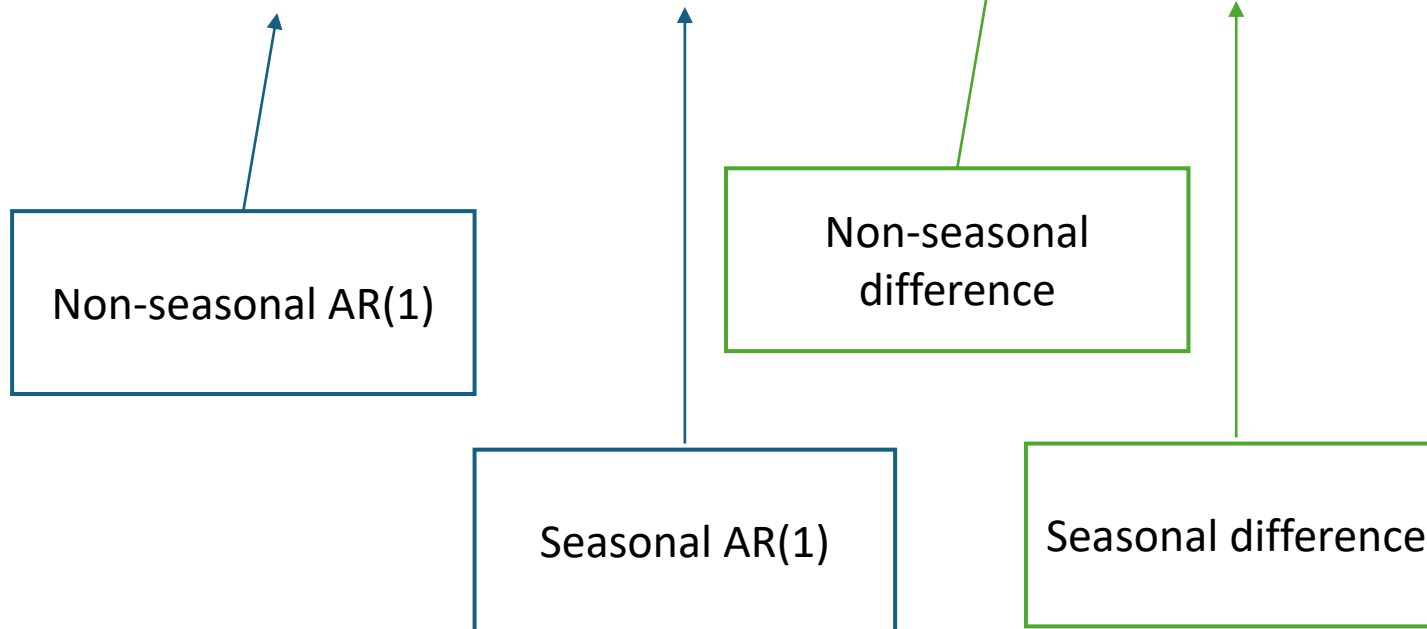
$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm})\epsilon_t$$

Seasonal ARIMA

Example: ARIMA(1,1,1)(1,1,1)₄

- Backshift notation

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = c + (1 + \theta_1 B)(1 + \Theta_1 B^4)\epsilon_t$$



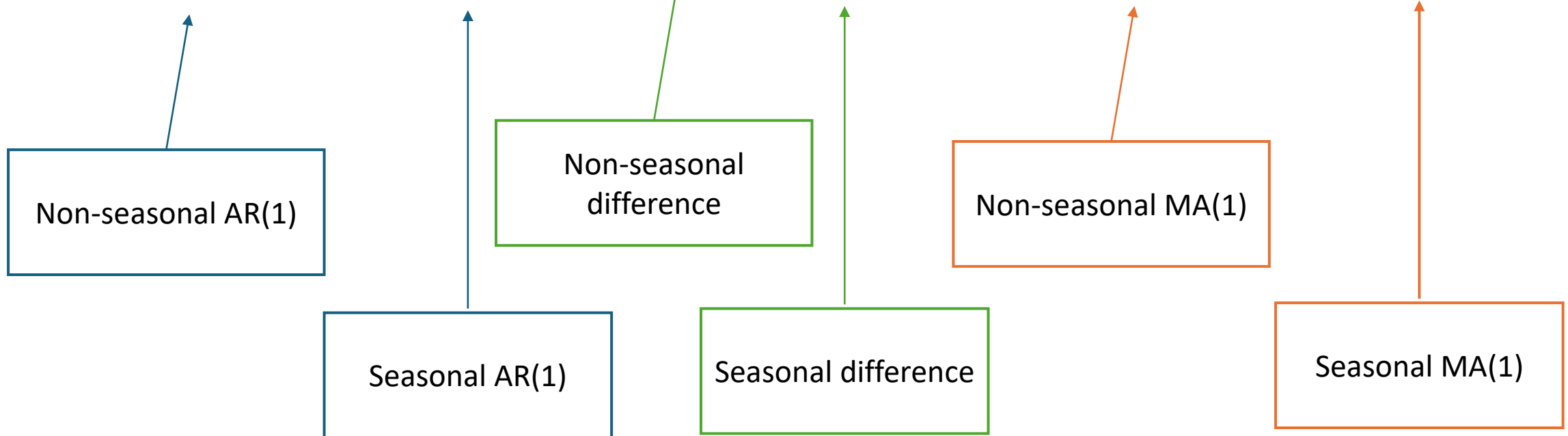
$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm})\epsilon_t$$

Seasonal ARIMA

Example: ARIMA(1,1,1)(1,1,1)₄

- Backshift notation

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = c + (1 + \theta_1 B)(1 + \Theta_1 B^4)\epsilon_t$$



Seasonal ARIMA

Example: ARIMA($\mathbf{1},1,1$)($\mathbf{1},1,1$)₄

- Backshift notation

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)y_t = c + (1 + \theta_1 B)(1 + \Theta_1 B^4)\epsilon_t$$

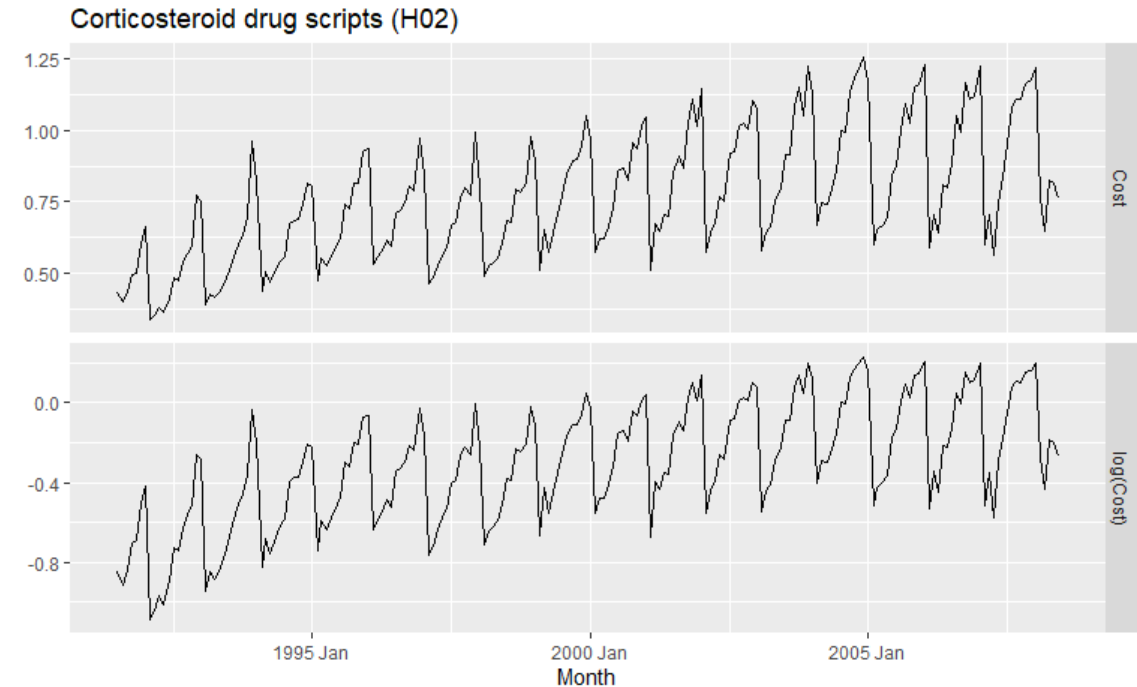
- All factors can be multiplied out and the general model written as

$$\begin{aligned} y_t = & c + (1 + \phi_1)y_{t-1} - \phi_1 y_{t-2} + (1 + \Phi_1)y_{t-4} \\ & - (1 + \phi_1 + \Phi_1 + \phi_1 \Phi_1)y_{t-5} + (\phi_1 + \phi_1 \Phi_1)y_{t-6} \\ & - \Phi_1 y_{t-8} + (\Phi_1 + \phi_1 \Phi_1)y_{t-9} - \phi_1 \Phi_1 y_{t-10} \\ & + \epsilon_t + \theta_1 \epsilon_{t-1} + \Theta_1 \epsilon_{t-4} + \theta_1 \Theta_1 \epsilon_{t-5} \end{aligned}$$

```
from sympy import *
phi1, Phi1, B = symbols('phi1 Phi1 B')
term_ar = 1 + phi1*B
term_sar = 1+Phi1*B**4
z1 = expand(term_ar*term_sar)
z1
[4] ✓ 0.0s
... B5Φ1φ1 + B4Φ1 + Bφ1 + 1
```

Example: Corticosteroid drug sales in Australia

```
h02 <- PBS %>%  
  filter(ATC2 == "H02") %>%  
  summarise(Cost = sum(Cost)/1e6)  
  
h02 %>%  
  mutate(log(Cost)) %>%  
  pivot_longer(-Month) %>%  
  ggplot(aes(x = Month, y = value)) +  
  geom_line() +  
  facet_grid(name ~ ., scales = "free_y") +  
  labs(y="", title="Corticosteroid drug scripts (H02)")
```



There is a small increase in the variance with the level, so we take logarithms to stabilize the variance.

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm}) \epsilon_t$$

Example: Corticosteroid drug sales in Australia (Con.)

```
h02 <- PBS %>%
  filter(ATC2 == "H02") %>%
  summarise(Cost = sum(Cost)/1e6)

fit <- h02 %>% model(auto = ARIMA(log(Cost)))
report(fit)
```

```
> report(fit)
Series: Cost
Model: ARIMA(2,1,0)(0,1,1)[12]
Transformation: log(Cost)

Coefficients:
      ar1      ar2      sma1
-0.8491  -0.4207  -0.6401
  φ1      φ2      Θ1
```

We can write the model using the backshift notation as follows:

$$(1 - \phi_1 B - \phi_2 B^2)(1 - B)(1 - B^{12})y_t = c + (1 + \Theta_1 B^{12})\epsilon_t$$

$$(1 + 0.8491B + 0.4207B^2)(1 - B)(1 - B^{12})y_t = (1 - 0.6401B^{12})\epsilon_t$$

Note the + and - signs

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - \Phi_1 B^m - \dots - \Phi_P B^{Pm})(1 - B)^d(1 - B^m)^D y_t = c + (1 + \theta_1 B + \dots + \theta_q B^q)(1 + \Theta_1 B^m + \dots + \Theta_Q B^{Qm}) \epsilon_t$$

In-class exercise

NZ arrivals

```
Series: Arrivals
Model: ARIMA(2,0,0)(1,1,2)[4] w/ drift

Coefficients:
      ar1      ar2      sar1      sma1      sma2      constant
    0.5830  0.1772 -0.7022  0.2447 -0.5612  3091.1032
s.e.  0.0898  0.0901  0.1866  0.1736  0.0964  762.7263

sigma^2 estimated as 1.51e+08:  log likelihood=-1331.03
AIC=2676.07  AICc=2677.04  BIC=2695.75
```

```
fit <- nzarrivals %>%
  model(arima = ARIMA(Arrivals))

report(fit)
```

Write the model using backshift notation.

Applications

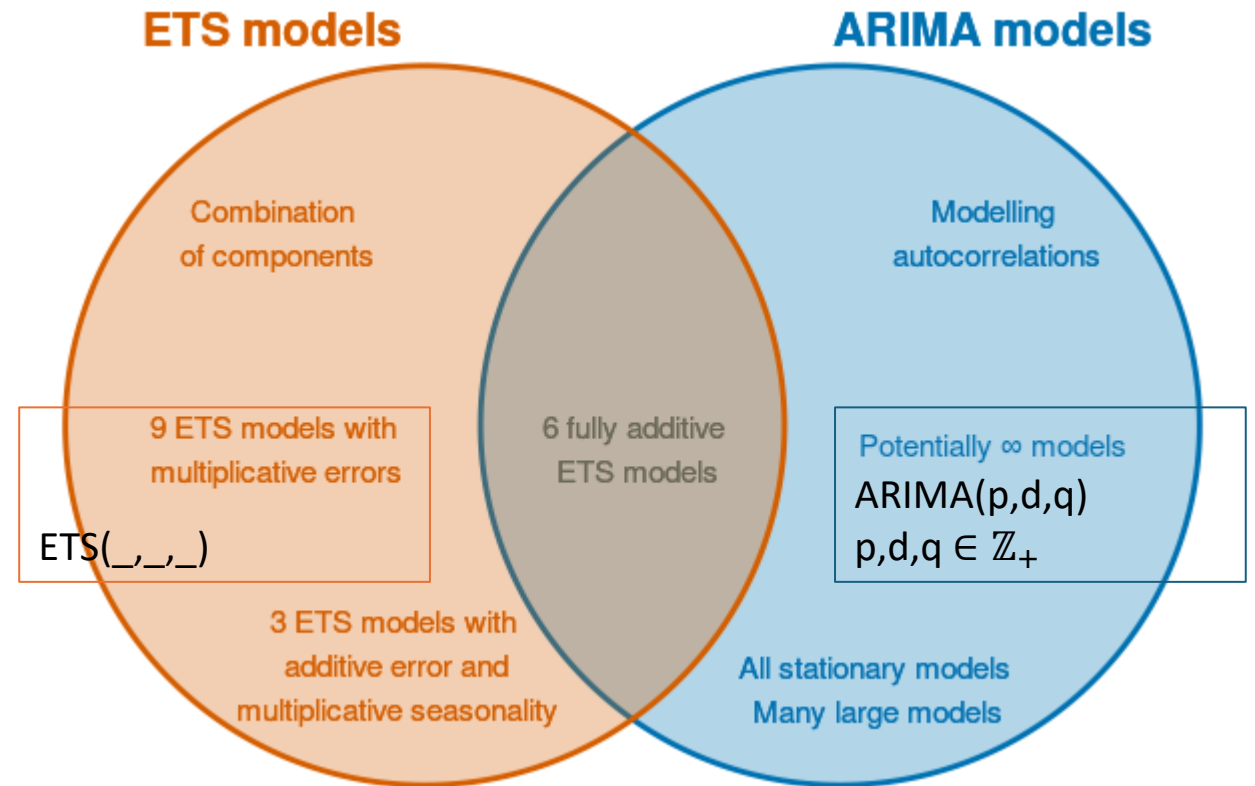
ARIMA vs ETS

Forecasting population

Applications in warehouse location

ARIMA vs ETS

- ETS and ARIMA models are the two most widely used approaches to TS forecasting.
- Exponential smoothing models are based on a description of the trend and seasonality in the data
- ARIMA models aim to describe the **autocorrelations** in the data



- While linear exponential smoothing models are all special cases of ARIMA models, the non-linear exponential smoothing models have no equivalent ARIMA counterparts.
- On the other hand, there are also many ARIMA models that have no exponential smoothing counterparts.
- In particular, all ETS models are non-stationary, while some ARIMA models are stationary

ARIMA vs ETS

Table 9.4: Equivalence relationships between ETS and ARIMA models.

ETS model	ARIMA model	Parameters
ETS(A,N,N)	ARIMA(0,1,1) Non-seasonal ARIMA	$\theta_1 = \alpha - 1$
ETS(A,A,N)	ARIMA(0,2,2)	$\theta_1 = \alpha + \beta - 2$ $\theta_2 = 1 - \alpha$
ETS(A,A _d ,N)	ARIMA(1,1,2)	$\phi_1 = \phi$ $\theta_1 = \alpha + \phi\beta - 1 - \phi$ $\theta_2 = (1 - \alpha)\phi$
ETS(A,N,A)	ARIMA(0,1,m)(0,1,0) _m Seasonal ARIMA (SARIMA)	
ETS(A,A,A)	ARIMA(0,1,m + 1)(0,1,0) _m	
ETS(A,A _d ,A)	ARIMA(1,0,m + 1)(0,1,0) _m	

Example AUS population: Comparing ARIMA and ETS via Cross-validation

```
aus_economy <- global_economy %>%
  filter(Code == "AUS") %>%
  mutate(Population = Population/1e6)

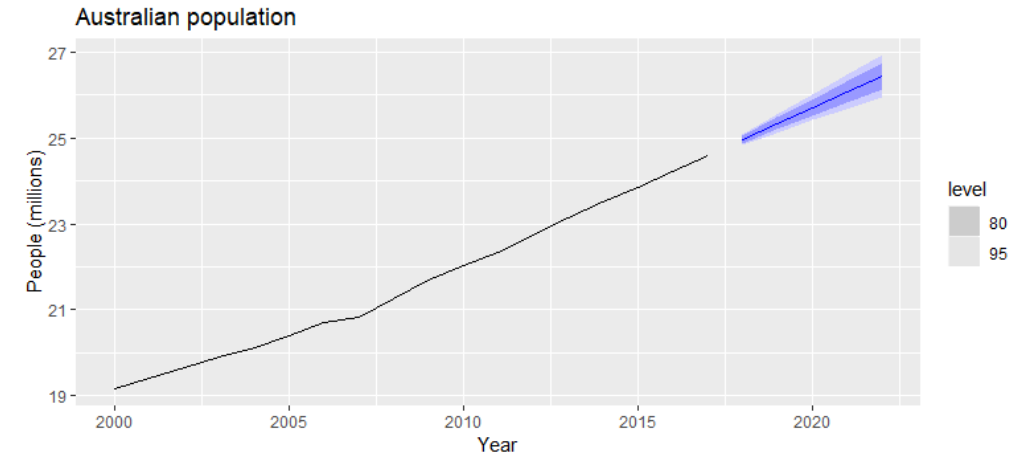
aus_economy %>%
  slice(-n()) %>%
  stretch_tsibble(.init = 10) %>%
  model(
    ETS(Population),
    ARIMA(Population)
  ) %>%
  forecast(h = 1) %>%
  accuracy(aus_economy) %>%
  select(.model, RMSE:MAPE)

#> # A tibble: 2 x 5
#>   .model      RMSE    MAE    MPE    MAPE
#>   <chr>      <dbl>  <dbl> <dbl> <dbl>
#> 1 ARIMA(Population) 0.194  0.0789 0.277 0.509
#> 2 ETS(Population)  0.0774 0.0543 0.112 0.327
```

In this case the ETS model has higher accuracy on the cross-validated performance measures.

We generate and plot forecasts for the next 5 years generated from an ETS model.

```
aus_economy %>%
  model(ETS(Population)) %>%
  forecast(h = "5 years") %>%
  autoplot(aus_economy %>% filter(Year >= 2000)) +
  labs(title = "Australian population",
        y = "People (millions)")
```



- Point-forecast metrics are appropriate when the forecast is used as a single planning input, such as expected demand, expected workload, expected revenue, or expected shipment volume.
- They are less appropriate when the decision depends directly on tail risk, service probability, shortage amount, or the full uncertainty distribution.

Business Insights: Why would a logistics manager care about population forecast accuracy?

- Population is often used as a proxy for future demand.
- A better point forecast gives better planning inputs for delivery volume, facility sizing, fleet planning, and capacity expansion.
- Examples:
 - Logistics firms (DHL, Kerry Express) care about urban population growth to size last-mile delivery networks.
 - Airlines may use population growth, together with income and tourism indicators, to plan route capacity.

From population forecasting to warehouse location

Where to rent a one-year warehouse?

Step 1: Forecast stage

- Forecast population for each country.
- Translate population into shipment demand (using simple multipliers, e.g., shipments $\propto$ population).

Step 2: Optimization stage

- Plug shipment volumes into WH location problem
- Show how the optimal warehouse shifts depending on demand balance.

Step 3: Business insight

- Firms deciding whether to place regional hubs in Bangkok vs HCMC often use this exact logic.



Node	Lon	Lat	Population	Shipments
Lao PDR	103	18.0		
Myanmar	96.2	16.8		
Thailand	101	13.8		
Vietnam	107	10.8		

Population $\rightarrow$ shipments $\rightarrow$ hub location.

Biased demand forecasts can shift the chosen hub location and create months of extra transport cost.

From population forecasting to WH location

Step 1: Forecasting

Approaches to multi-country forecasting

- Country-specific model
 - Choose the best model for each country using cross-validation.
 - Example: ARIMA for Thailand, ETS for Vietnam.
- Same model for all countries
 - Easier to explain and maintain.
 - May be chosen for organizational consistency.
- Forecast combination
 - Average or weighted average of several models.
 - Often robust when no single model dominates.
- Business-rule adjustment
 - Override or adjust statistical forecasts when there is known market information.
 - Example: new market, policy change, GDP growth, special campaign.

.model	Country	ME	RMSE	MAE	MPE	MAPE	MASE
<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
ARIMA(Population)	Lao PDR	0.0677	0.185	0.0679	1.08	1.08	0.820
ARIMA(Population)	Myanmar	0.0553	0.146	0.0604	0.110	0.120	0.106
ARIMA(Population)	Thailand	0.261	0.745	0.267	0.389	0.397	0.360
ARIMA(Population)	Vietnam	0.611	1.61	0.614	0.689	0.692	0.555
ETS(Population)	Lao PDR	0.000510	0.00585	0.00563	0.00534	0.0859	0.0679
ETS(Population)	Myanmar	0.0351	0.0435	0.0351	0.0688	0.0688	0.0616
ETS(Population)	Thailand	0.00217	0.0189	0.0170	0.00342	0.0249	0.0229
ETS(Population)	Vietnam	0.0539	0.109	0.0731	0.0609	0.0811	0.0661

```
#####  
# Application  
# Warehouse location  
#####  
  
df <- global_economy %>%  
  filter(Code %in% c('THA', 'VNM', 'MMR', 'LAO')) %>%  
  mutate(Population = Population/1e6)  
  
accTable <- df %>%  
  filter(Year >= 2000 & Year < 2017) %>%  
  stretch_tsibble(.init = 10) %>% # Try .init = 3. ETS fails  
  model(  
    ETS(Population),  
    ARIMA(Population)  
  ) %>%  
  forecast(h = 1) %>%  
  accuracy(df)  
  
accTable %>%  
  select(c(.model, Country, ME:MASE))  
  
accTable %>%  
  group_by(.model) %>%  
  summarise(avg_MAE = mean(MAE, na.rm = TRUE)) %>%  
  arrange(avg_MAE)
```

.model	avg_MAE
<chr>	<dbl>
ETS(Population)	0.0327
ARIMA(Population)	0.252

From population forecasting to warehouse location

Step 2: WH Location problem

Input parameters

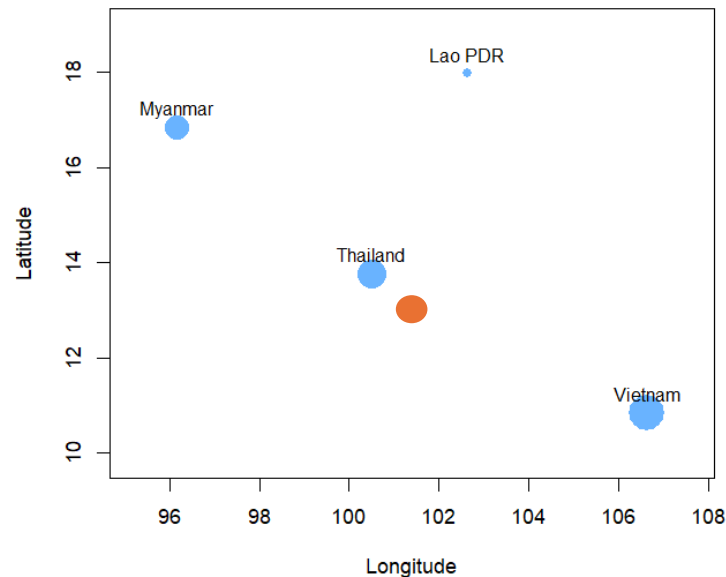
r_j = forecasted annual shipment volume for country j
 (x_j, y_j) = location of the demand

Decision variable

$\mathbf{v} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$ location of the warehouse

Minimize

$$\sum_{j=1}^4 r_j \|\mathbf{u}_j - \mathbf{v}\|_2 = \sum_{j=1}^4 r_j \sqrt{(x_j - x_0)^2 + (y_j - y_0)^2}$$



Multiplier	10%					
Customer	Country	Long	Lat	Distance	PopFC2018	Annual shipments
1	Lao PDR	103	18	4.65	6.96	0.696
2	Myanmar	96.2	16.8	5.66	53.9	5.39
3	Thailand	101	13.8	0.00	69.2	6.92
4	Vietnam	107	10.8	6.71	96.5	9.65
WH location		101.00	13.80			
Total yearly distance from WH to customers						98.48

Discussion & in-class exercise

- If node 1 volume doubles from heavy advertising, where would you move the warehouse?
- What if the contract is three years?

Applications in logistics: Summary

- Earlier, we used quantile score when the decision was a stock level for a target service probability.
- We used continuous-rank probability score (CRPS) when the whole predictive distribution mattered.
- But not every business decision needs the full distribution.
- Sometimes the forecast is used as a single planning number.
- For example, warehouse-location model. Point-forecast accuracy (e.g., MAE, RMSE, MAPE, MASE) is directly relevant.
- The right error measure depends on how the forecast will be used, i.e., the decision target.

Quick check: In each of these decision problem, choose the appropriate error measure among MAE, QS (quantile score), CRPS.

1. Warehouse-location problem: MAE
2. Stock level given type-1 (in-stock probability) service level: QS
3. Stock level given type-2 (fill rate) service level: CRPS
4. Forecast monthly shipment volume for annual budget planning: _____
5. Forecast delivery time to set a promise time that is met 90% of the time: _____
6. Forecast hourly call/order arrivals to plan average workload: _____ In queueing models, we often need an arrival rate, such as λ .
7. VaR: Forecast warehouse monthly electricity cost to set a budget that is exceeded only 10% of the time: _____
8. CVaR: Forecast warehouse monthly electricity cost to estimate the average cost in unusually high-cost months: _____

Homework

Homework

Write out the model using backshift notation.

- ARIMA(0,0,1)(0,0,1)₁₂

$$y_t = c + (1 + \theta_1 B)(1 + \Theta_1 B^{12})\epsilon_t$$

- ARIMA(1,0,0)(1,0,0)₁₂

- ARIMA(0,0,0)(0,1,1)₁₂

- ARIMA(1,1,0)(0,1,2)₄

Appendix

Backshift notation $B^m y_t = y_{t-m}$

- $By_t = y_{t-1}$
- $B^2 y_t = y_{t-2}$
- $y_t - y_{t-1} = y_t - By_t = (1 - B)y_t := y'_t$ **the first difference**
- $y_t - y_{t-2} = (1 - B^2)y_t$ the **second difference**

- A **seasonal difference** is the difference between an observation and the corresponding observation from the previous year.

$$(1 - B^m)y_t$$

- Quarterly: seasonal period $m = 4$.
Seasonal difference $y_t - y_{t-4} = (1 - B^4)y_t$

- Monthly:
Seasonal difference

- A **seasonal difference** followed by a first difference can be written as

$$(1 - B)(1 - B^m)y_t$$

The **second-order difference**

$$\begin{aligned} y''_t &= y'_t - y'_{t-1} = y_t - 2y_{t-1} + y_{t-2} \\ &= (1 - 2B + B^2)y_t = (1 - B)^2 y_t \end{aligned}$$

This is not the same as the second difference $(1 - B^2)y_t$

The backshift operator B can be treated using ordinary algebraic rules.

$$\begin{aligned} (1 - B)(1 - B^m)y_t &= (1 - B - B^m + B^{m+1})y_t \\ &= y_t - y_{t-1} - y_{t-m} + y_{t-m-1} \end{aligned}$$

Multiplying polynomials

- A **polynomial** is a mathematical expression consisting of variables) and coefficients, that involves only the operations of addition, subtraction, multiplication, and positive-integer powers of variables
 - Single variable, e.g., $x^2 - 4x + 7$
 - Standard form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$
 - Three variables, e.g., $x^3 - 2xyz^2 - yz + 1$
- Multiplying polynomials
 - https://www.youtube.com/watch?v=yJzLYa-_Y1k



Multiplying polynomials example | Polynomial and rational functions | Algebra II | Khan Academy

Fundraiser

Khan Academy
8.26M subscribers

Subscribe

455

Share

Download

...

Greek alphabet

Α α alpha	Β β beta	Γ γ gamma	Δ δ delta	Ε ε epsilon
Ζ ζ zeta	Η η eta	Θ θ theta	Ι ι iota	Κ κ kappa
Λ λ lambda	Μ μ mu	Ν ν nu	Ξ ξ xi	Ο ο omikron
Π π pi	Ρ ρ rho	Σ σ/ς sigma	Τ τ tau	Υ υ upsilon
Φ φ phi	Χ χ chi	Ψ ψ psi	Ω ω omega	

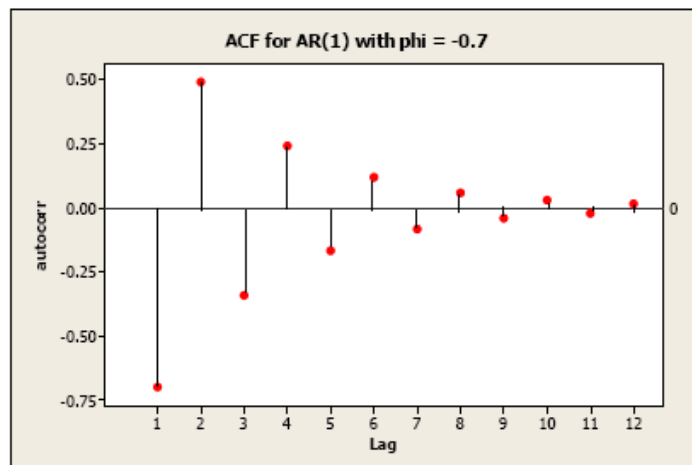
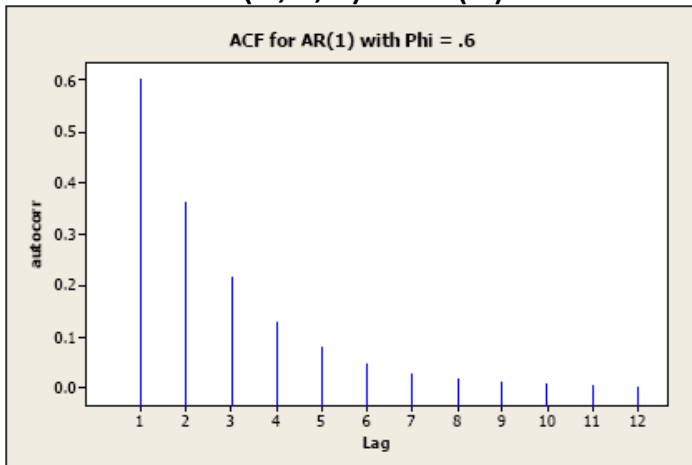
ARIMA Theory*

ARIMA Overview

Auto**r**egressive **i**ntegrated **m**oving **a**verage: **ARIMA**(**p**,**d**,**q**)

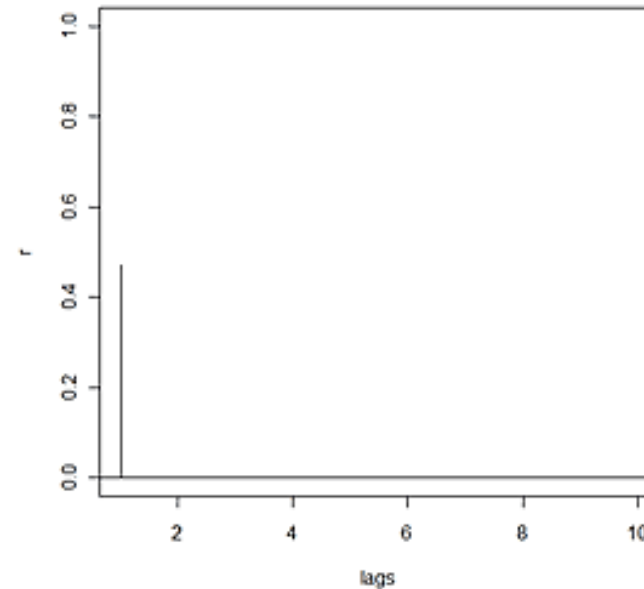
AR process, ACF dies out

ARIMA(1,0,0)= AR(1)



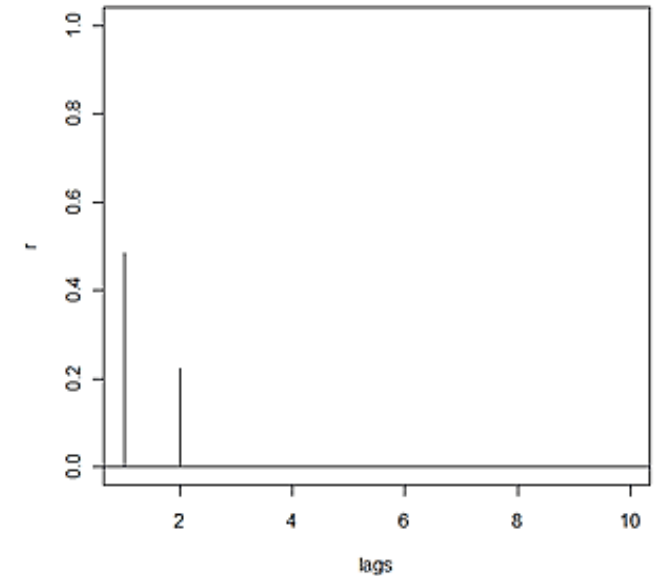
MA(q) process, ACF cut off after q spikes

ACF for MA(1) with theta1 = 0.7



ARIMA(0,0,1) = MA(1)

ACF for MA(2) with theta1 = 0.5, theta2=0.3



ARIMA(0,0,2) = MA(2)

ARIMA captures patterns (via **autocorrelations**) in TS by combining three ingredients:

AR (past values)

I (differences)

MA (past shocks)

AR order $p \in \mathbb{Z}_+$

d th differencing, usually 0,1,2

MA order $q \in \mathbb{Z}_+$

Recall #04: **ACF of white noise** no structure $\rightarrow$ ARIMA(0,0,0)

Review: Model error assumption

Random error term ϵ_t

- Assume

1. [*zero bias*] Expected value of each error term is zero: $E[\epsilon_t] = 0$

2. [**white noise**] Errors for different time periods are independent of (or at least uncorrelated with) one another and also independent of past states

No autocorrelation
 $corr(\epsilon_t, \epsilon_{t-k}) = 0$

3. [*homoskedasticity*] Variance of the errors is constant: $var(\epsilon_t) = \sigma^2$

4. Errors are drawn from a normal distribution

- A short-hand notation $\epsilon_t \sim \text{NID}(0, \sigma^2)$

- NID stands for “normally and independently distributed”.

Portmanteau tests for autocorrelation

- Let $\rho_j = \text{corr}(\epsilon_t, \epsilon_{t-j})$ be the autocorrelation at lag j
- Consider a *whole* set of r_k values, and develop a test to see whether the set is significantly different from a zero set.

$$H_0: \rho_1 = \rho_2 = \dots = \rho_\ell = 0$$

- Let ℓ be the maximum lag being considered
 - We suggest using $\ell = 10$ for non-seasonal data and $\ell = 2m$ for seasonal data, where m is the period of seasonality.
- Box-Pierce test** $Q = T \sum_{k=1}^{\ell} r_k^2$
- Ljung-Box test** $Q^* = T(T+2) \sum_{k=1}^{\ell} (T-k)^{-1} r_k^2$
- If the autocorrelations did come from a white noise series, then both Q and Q^* would have a χ^2 distribution with $(\ell - K)$ degrees of freedom, where K is the number of parameters in the model.
 - Naïve method has no parameters, so $K = 0$
 - Drift method has one estimated parameter (drift coefficient), so $K = 1$

```
aug %>% features(.innov, box_pierce, lag = 10, dof = 0)
#> # A tibble: 1 × 4
#>   Symbol .model      bp_stat bp_pvalue
#>   <chr>   <chr>      <dbl>   <dbl>
#> 1 GOOG  NAIVE(Close)    7.74    0.654
```

```
aug %>% features(.innov, ljung_box, lag = 10, dof = 0)
#> # A tibble: 1 × 4
#>   Symbol .model      lb_stat lb_pvalue
#>   <chr>   <chr>      <dbl>   <dbl>
#> 1 GOOG  NAIVE(Close)    7.91    0.637
```

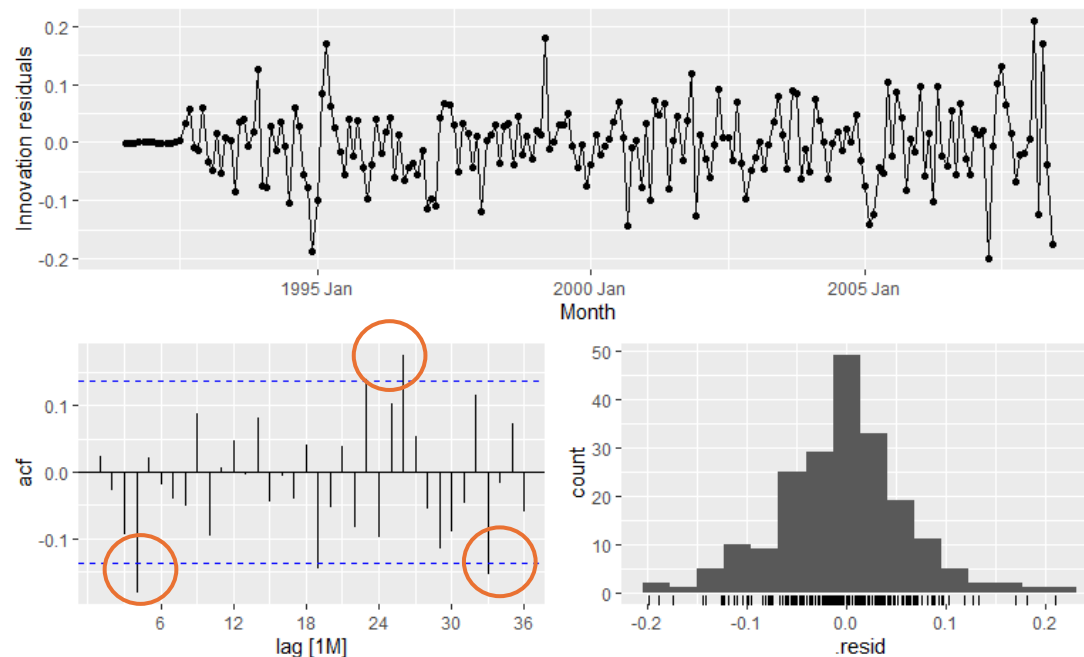
If the P-value is less than (or equal to) the significance level α , then there is sufficient evidence to reject the null hypothesis: the null hypothesis is rejected in favor of the alternative hypothesis.

Example: Corticosteroid drug sales in Australia (Con.)

```
fit <- h02 %>% model(auto = ARIMA(log(Cost)))
report(fit)

fit %>% gg_tsresiduals(lag_max=36)
augment(fit) %>%
  features(.innov, ljung_box, lag = 36)
```

	.model	lb_stat	lb_pvalue
	<chr>	<dbl>	<dbl>
1	auto	59.3	0.00859



- There are a few **significant spikes** in the ACF, and the model fails the Ljung-Box test.
- The model can still be used for forecasting.
- However, the prediction intervals may not be accurate due to the correlated residuals.
 - Value at risk (VaR) → 95% percentile of forecast distribution
- Sometimes it is just not possible to find a model that passes all of the tests.

Model selection

- Minimizing AIC, AICc or BIC.
- Textbook's preference: AICc
- Note that AIC values between models with different orders of differencing are not comparable.

Hyndman-Khandakar algorithm for automatic ARIMA modelling

1. The number of differences $0 \leq d \leq 2$ is determined using repeated KPSS tests.
2. The values of p and q are then chosen by minimising the AICc after differencing the data d times. Rather than considering every possible combination of p and q , the algorithm uses a stepwise search to traverse the model space.
 - a. Four initial models are fitted:
 - $\text{ARIMA}(0, d, 0)$,
 - $\text{ARIMA}(2, d, 2)$,
 - $\text{ARIMA}(1, d, 0)$,
 - $\text{ARIMA}(0, d, 1)$.A constant is included unless $d = 2$. If $d \leq 1$, an additional model is also fitted:
 - $\text{ARIMA}(0, d, 0)$ without a constant.
 - b. The best model (with the smallest AICc value) fitted in step (a) is set to be the "current model".
 - c. Variations on the current model are considered:
 - vary p and/or q from the current model by ± 1 ;
 - include/exclude c from the current model.The best model considered so far (either the current model or one of these variations) becomes the new current model.
 - d. Repeat Step 2(c) until no lower AICc can be found.

